

Semantic Routing in NDN Networks

Utilizing Vector Embeddings for Concept-Based Forwarding



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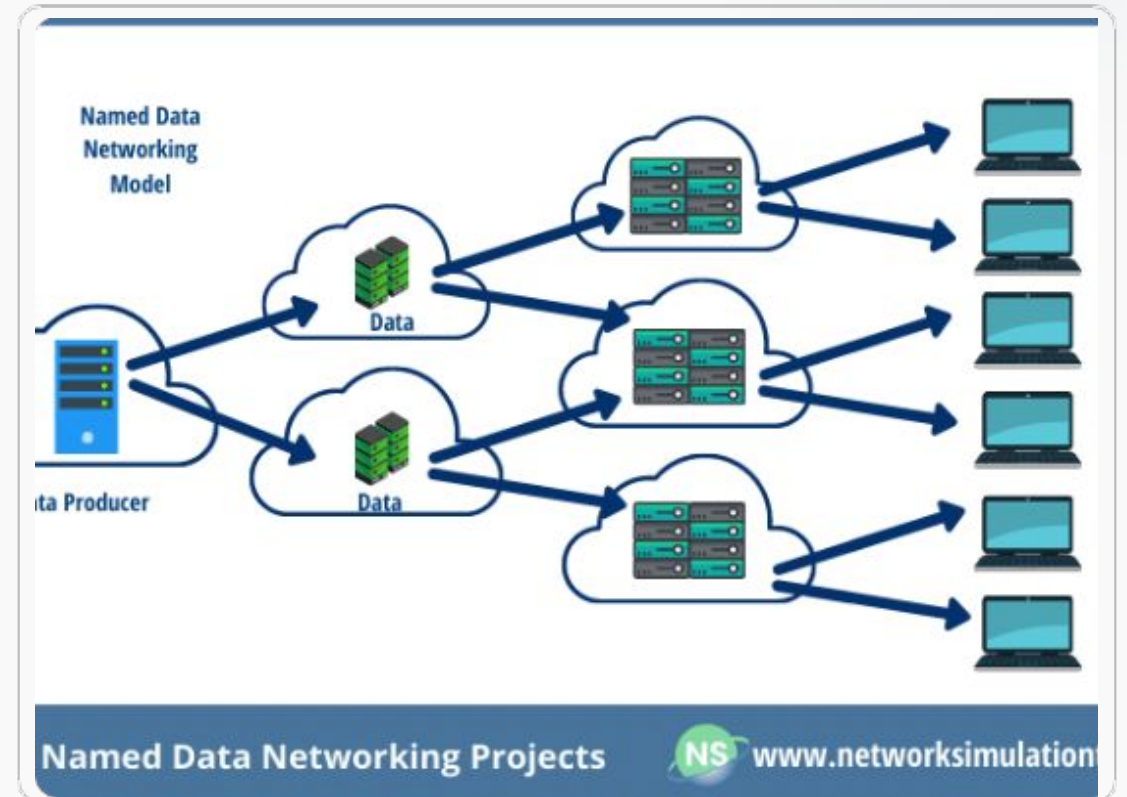
Distributed Systems Project

The Bottleneck: Exact String Matching

NDN (Named Data Networking) replaces IP addresses with content names. However, current routers rely on **Exact String Matching** in FIB tables.

The Failure Point

If a user requests `/dist-sys/hw1` but the cache stores `/distributed-systems/homework1`, the network fails to recognize the semantic equivalence, leading to a **Cache Miss**.



Semantic Routing Approach



AI-Driven Layer

Integrating Lightweight LLM models directly at the network edge to interpret intent.



Vector Embeddings

Transforming text-based names into high-dimensional numerical vectors.



Similarity Search

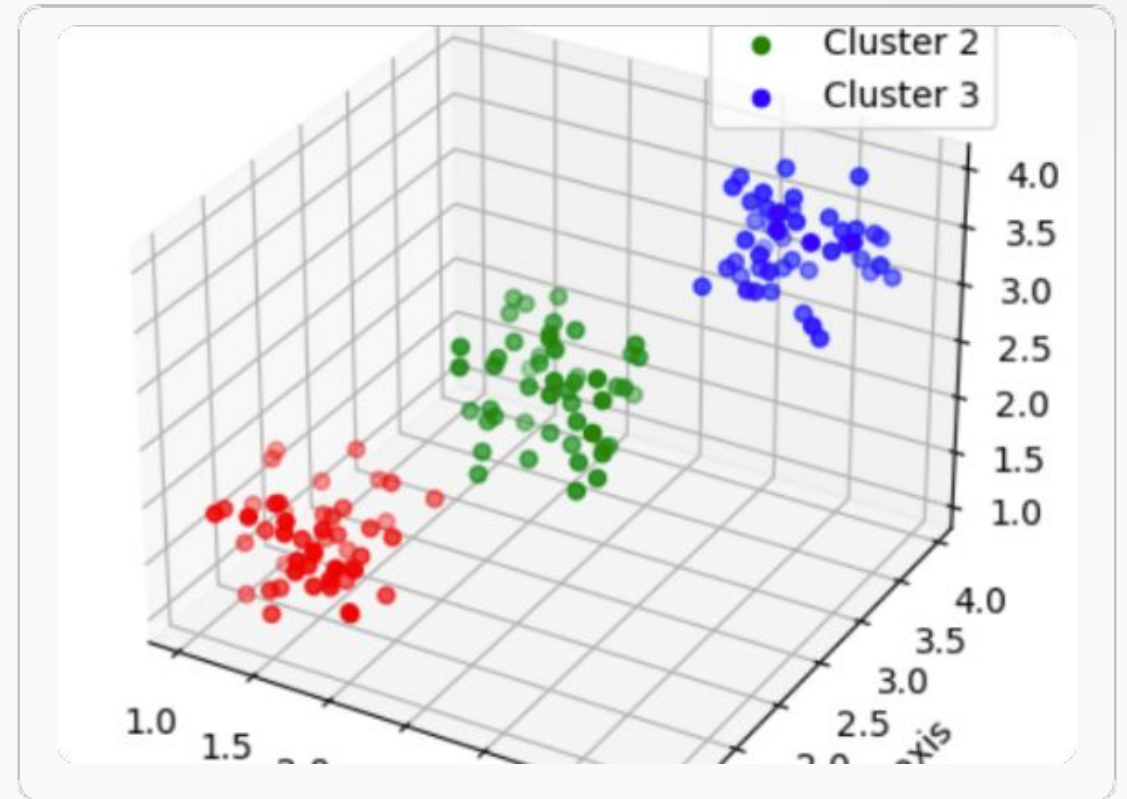
Forwarding based on mathematical distance rather than character identity.

Vector Space & Cosine Similarity

By mapping names into a semantic space, related concepts cluster together. We measure the angle between vectors A and B.

$$\text{Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

Where a value of 1 indicates semantic identity and 0 indicates complete irrelevance.



100



1. Packet Ingress: Interest packet is intercepted by the router.

2. Embedding: The name is processed by a Sentence-Transformer (e.g., MiniLM).

3. Vector Search:

FIB engine uses **FAISS** for ultra-fast neighbor lookup.

4. Decision: Forwarded if similarity exceeds a defined Threshold.

Distributed System Aspects



No Central Master

Every router maintains its own Local Vector Index, ensuring zero single-point-of-failure.



Eventual Consistency

Semantic tables are updated across the network using peer-to-peer protocols.



Overhead Mgt.

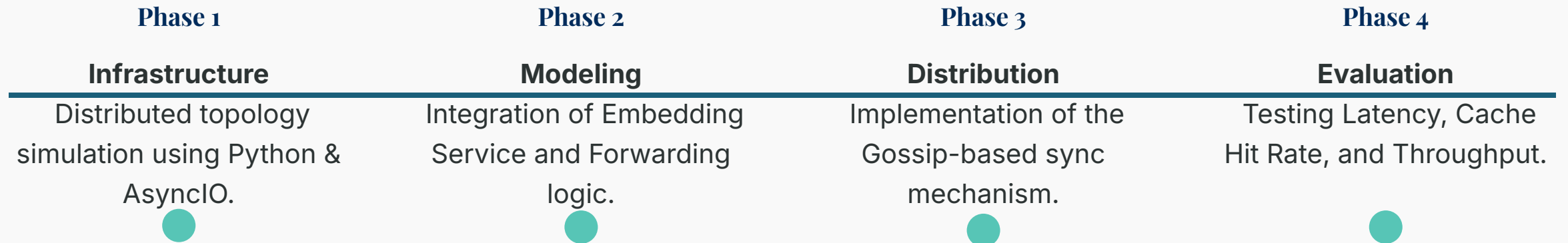
Balancing embedding latency with networking line-speed via quantization.

Synchronization via Gossip

Routers exchange newly learned semantic relationships with neighbors at random intervals.

- **Anti-Entropy:** Comparing vector signatures to find mismatches.
- **Rumor Spreading:** Rapid propagation of high-value metadata.
- **Network Friendly:** Only "Delta" updates are sent to save bandwidth.

Implementation Phases



Performance & Search Speed

< 2ms

Search Latency

Why FAISS?

Developed by Meta, FAISS allows for **Approximate Nearest Neighbor (ANN)** search. Even with millions of content names, it finds the best match in sub-millisecond timeframes, preventing the router from becoming a network bottleneck.

NDN vs. Semantic NDN

Feature	Standard NDN	Semantic NDN (Proposed)
Matching Method	Exact String	Vector Similarity
Resilience to Typos	Zero (Packet Drop)	High (Nearest Match)
Cache Efficiency	Moderate	Significantly Increased
Compute Overhead	Minimal	Medium (Edge GPU/NPU required)

Key Takeaways

- **Intelligence:** Moving from "Dumb" routers to "Semantic Aware" nodes.
- **Efficiency:** Reducing redundant data traffic by resolving name mismatches.
- **Distributed Core:** Gossip protocols ensure the network evolves and learns together.

"The future of NDN lies not just in routing names, but in routing meanings."

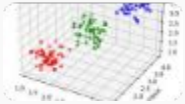
Thank You!

Image Sources



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Source: networksimulationtools.com



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